



**CLEAN DEVELOPMENT MECHANISM
PROJECT DESIGN DOCUMENT FORM (CDM-PDD)
Version 03 - in effect as of: 28 July 2006**

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**SECTION A. General description of project activity****A.1 Title of the project activity:**

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Guangdong Nan Ao 26MW Wind Power Project

Version number of the document: 4

Date: 12/03/2008

Revision history of PDD

Version Number	Date
Version 1	19/12/2006
Version 2	28/11/2007
Version 3	28/02/2008
Version 4	12/03/2008

A.2. Description of the project activity:

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The Guangdong Nan Ao 26MW Wind Power Project* (hereafter refers to the proposed project) is to utilize wind resources for electricity generation through the construction of a wind farm with a total capacity of 23.25MW and a 110kV substation at Nan Ao Island, Nan Ao Country, Shantou City, Guangdong Province, P. R. China. The electricity generated from the project will be sold to the Guangdong Provincial Power Grid, an integral part of the Southern China Power Grid. The estimated annual net electricity output and emission reductions of the proposed project are 57,595 MWh and 53,425 tCO₂e, respectively. The proposed project is a grid-connected renewable energy project.

The proposed project makes contribution to the local sustainable development as follows:

1. GHG emission reduction

The project is designed to produce 57,595MWh of electricity per year, from which the electricity generated will be sold to the Guangdong Provincial Power Grid, an integral part of the China Southern Power Grid. The proposed project activity will achieve obvious greenhouse gas (GHG) emission reductions by avoiding CO₂ emissions, as grid-connected fossil fuel-fired power plants are dominated in the China Southern Power Grid. The annual average estimated emission reduction over the first crediting period is 53,425 t CO₂e.

2. Pollutants emission reduction through replacing fossil fuel combustion

The proposed project is to replace grid-connected fossil fuel-fired power plants in the China Southern Power Grid, and thus reduce fossil fuel consumption and avoid pollutants emission, such as sulfur dioxide and dust, brought by fossil fuel combustion. Therefore, the proposed project has obvious environmental benefit.

3. Employment opportunities

The conducting of the proposed project will offer about 50 job opportunities for local people, and achieve the economic growth in the region. Furthermore, the proposed project plans to utilize domestic made

* Guangdong Development Reform Commission approved a total of 26MW for the proposed project as named "Guangdong Nan Ao 26MW Wind Power Project". However, due to the limit of microscopic site selection, three wind turbines (3*750kw) are not to be installed. Finally only 23.25MW of the proposed project will be installed.



state-of-the-art wind turbines, which will promote Chinese wind turbine manufacturing industry by increasing its market share and the employment opportunities.

A.3. Project participants:

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Please list project participants and Party(ies) involved and provide contact information in Annex 1. Information shall be indicated using the following tabular format.

Name of Party involved (*) (host indicates a host Party)	Private and/or public entity(ies) project participants (*) (as applicable)	Kindly indicate if the Party involved wishes to be considered as project participant (Yes/No)
Peoples' Republic of China (host)	China Resources Wind Power (Shantou) Co., Ltd	No
UK	Merrill Lynch Commodities Europe limited	No

A.4. Technical description of the project activity:**A.4.1. Location of the project activity:**

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A.4.1.1. Host Party(ies):

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People's Republic of China

A.4.1.2. Region/State/Province etc.:

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Guangdong Province

A.4.1.3. City/Town/Community etc:

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Nan Ao County, Shantou City

A.4.1.4. Detail of physical location, including information allowing the unique identification of this project activity (maximum one page):

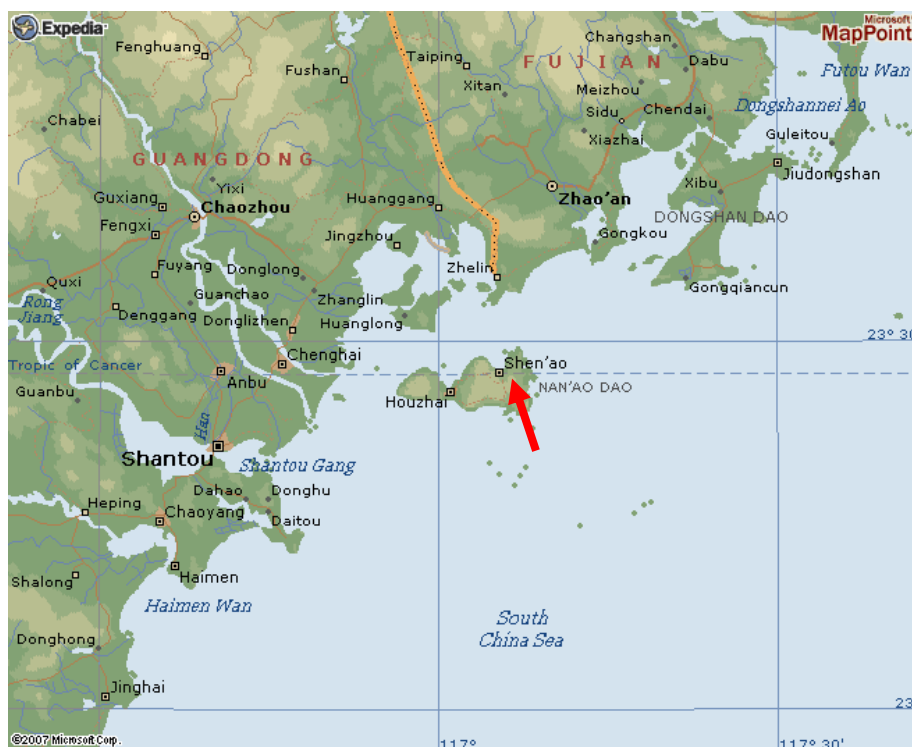
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The proposed project is located on Nan Ao Island, Nan Ao County, Shantou City, Guangdong Province, People's Republic of China. The project has geographical coordinates with east longitude of 117°5'27.73" and north latitude of 23°25'12.63". Figures A1, A2 show the geographical location of the proposed project.

Figure A1. The proposed project in the map of P. R. China



Figure A2. The proposed project in the map of Nan A'o Island



A.4.2. Category(ies) of project activity:

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Category: Renewable electricity in grid connected applications
Sectoral Scope: 1 Energy industries

A.4.3. Technology to be employed by the project activity:

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The proposed project involves the installation of 31 wind turbines with each capacity of 750 kW, and total up an installation capacity of 23.25MW. The annual operating hour of the proposed project is 2477. The wind turbines will be supplied by a domestic manufacturer – Xinjiang Goldwind Science and Technology Co., Ltd. The selected model is S48/750. The main technical parameters of the wind turbine are provided in the following table.

Item	Unit	Index
Rated capacity	kW	750
Number of blades		3
Rotor diameter	m	44
Swept area	m ²	2124
Startup speed	m/s	4
Rated wind speed	m/s	16
Shutoff speed	m/s	25
Rated voltage of generator	V	690
Power conditioner		Pitch regulated and variable speed
Control system		PC

The proposed project will also establish an 110kv substation on Nan Ao Island, through which the proposed project is connected to the 110KV Niutouling Substation in Nan Ao Island, and the generated electricity will be sold to the Guangdong Provincial Power Grid, which is an integral part of the China Southern Power Grid. The proposed project is to use the domestic wind turbines, therefore it doesn't involve any technical transfer.

A.4.4 Estimated amount of emission reductions over the chosen crediting period:

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A crediting period of 7 (2008.11-2015.10) years (renewable twice) is selected for the project activity. An estimation of emissions reductions expected over the crediting period is provided in the table below¹.

Years	Annual estimation of emission reductions in tonnes of CO ₂ e
2008(Nov.-Dec.)	8,904
2009	53,425
2010	53,425
2011	53,425
2012	53,425

¹The purchasing contract was countersigned between the equipment supplier and the project owner on 30th April 2007. The proposed project started construction on 10th June 2007 and the first wind turbine was installed on 20th October 2007. The first turbine is expected to be put into delivery on 31st December 2007 and all the turbines are expected to be put into delivery on 6th February 2008.



2013	53,425
2014	53,425
2015(Jan.-Oct.)	44,521
Total estimated reductions (tonnes of CO ₂ e)	373,975
Total number of the first crediting years	7
Annual average over the crediting period of estimated reductions (tonnes of CO₂e)	53,425

A.4.5. Public funding of the project activity:

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There is no public funding for this project.

**SECTION B. Application of a baseline and monitoring methodology****B.1. Title and reference of the approved baseline and monitoring methodology applied to the project activity:**

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The approved methodology applied in the proposed project activity is ACM0002 (the 6th version) – “Consolidated methodology for grid-connected electricity generation from renewable sources”. For more information regarding the methodology please refer to <http://cdm.unfccc.int/methodologies/approved>. Version 3 of “Tool for the Demonstration and Assessment of Additionality” is also applied in the proposed project.

B.2 Justification of the choice of the methodology and why it is applicable to the project activity:

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The proposed project meets all applicability conditions of methodology ACM0002 which are listed as follows:

- 1) The proposed project is a new wind power project;
- 2) The proposed project does not involve switching from fossil fuels to renewable energy at the site;
- 3) The geographic and system boundaries of China Southern Power Grid to which the proposed project will be connected can be clearly identified and information on the characteristics of the grid is available.

B.3. Description of the sources and gases included in the project boundary

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	Source	Gas	Included?	Justification / Explanation
Baseline	Fossil fuels fired power plants in the Southern China Power Grid	CO ₂	Yes	Major emission sources
		CH ₄	No	Excluded for simplification. This is conservative.
		N ₂ O	No	Excluded for simplification. This is conservative.
Project Activity	On-site fuel combustion to the project activity	CO ₂	No	Excluded for simplification.
		CH ₄	No	Excluded for simplification.
		N ₂ O	No	Excluded for simplification.

The spatial extent of the project boundary includes Guangdong Nan Ao 26MW Wind Power Project site. The project site includes the power plants, turbines themselves and auxiliary electric equipments that are used to support the turbines operation. The Guangdong Nan Ao 26MW Wind Power Project is connected to the Southern China Power Grid and the Southern China Power Grid including all power plants connected is selected as the project boundary therefore.

The Southern China Power Grid is the project electricity system, which is defined by the spatial extent of the power plants that can be dispatched without significant transmission constraints. In determining this project electricity system, such causes below are taken into account.



Using the boundary definitions published by the Chinese DNA, the Southern China Power Grid consists of Guangdong, Guangxi Zhuang Autonomous Region, Guizhou and Yunnan Provincial Power Grids. The electricity transmission between different provinces in the Southern China Power Grid is large. The electricity transmission among the Southern China Power Grid and the other grids are very insignificant^{1.1}. Hence, it is reasonable for the proposed project to regard the Southern China Power Grid as the project boundary.

In April 2002, China implemented the policy "Separate power plants from network and compete in price to enter network". The power sector reform started in 2002, and the five big power grids structured from then. Hence the grid data from Year 2003, 2004 and 2005 of the Southern China Power Grid applied in the PDD, which is after the power sector reform in China.

B.4. Description of how the baseline scenario is identified and description of the identified baseline scenario:

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According to ACM0002 ver06, for project activities that do not modify or retrofit an existing electricity generation facility, the baseline is the following:

Electricity delivered to the grid by the project would have otherwise been generated by the operation of grid-connected power plants and by the addition of new generation sources, as reflected in the combined margin (CM) calculations described below.

Therefore, since the proposed project activity does not modify or retrofit an existing electricity generation facility, supply of equivalent annual power output by the Southern China Power Grid to which the proposed project is connected is the baseline scenario for the proposed project activity.

B.5. Description of how the anthropogenic emissions of GHG by sources are reduced below those that would have occurred in the absence of the registered CDM project activity (assessment and demonstration of additionality): >>

The additionality of the proposed project is demonstrated and assessed by the approved Tool for the Demonstration and Assessment of Additionality. Following steps include:

Step 1. Identification of alternatives to the project activity consistent with current laws and regulations

Define realistic and credible alternatives to the project activity(s) that can be (part of) the baseline scenario through the following sub-steps:

Sub-step 1a. Define alternatives to the project activity:

In absence of the proposed project, reasonable and credible alternatives that are in accordance with current laws and regulations include:

- 1) The proposed project not taken as CDM project;
- 2) Construction of a fossil fuel power plant with equivalent amount of annual electricity output;

^{1.1} The total power generation of the Southern China Power Grid in 2005 is 4209.5611GWh. The net import from the other power grid to the Southern China Power Grid in 2005 is only 112.2982GWh (total import is 246.1158GWh and total output is 133.8176GWh), which is only 2.6% of the total generation. Source: China Electric Power Yearbook 2006



- 3) Construction of a power plant using other source of renewable equivalent amount of annual electricity output; and
- 4) Supply of equivalent annual power output by the China Southern Power Grid where the proposed project is connected.

In conclusion, the only practical and feasible baseline scenario is the alternative 4).

Sub-step 1b. Consistency with mandatory laws and regulations:

The alternative fossil fuel power plant with the equivalent power output as the proposed project will be less than 135MW, and considering China's regulations of energy conservation for electric power industry, coal-fired plants with a capacity of 135MW or less are prohibited from building in large grid such as provincial grids² and the fossil fuel-fired power units with less than 100MW capacity is strictly regulated for installation³ in China, therefore, the Alternative 2), is not a realistic and acceptable alternative.

The alternative 1), 3) and 4) are in line with the existing Chinese laws and regulations.

However, for the alternative 3), besides wind power, solar PV, geothermal, biomass and hydro are the possible grid-connected renewable energy technologies that could be applied in the China Southern Power Grid. However, there is no other renewable primary energy in the Nan Ao Island except a minimal quantity of hydro generators⁴. Due to the fact that Nan Ao County is a small island and lacks of usable hydro resources⁵, it is not suitable to develop a hydropower plant with similar capacity around the site of the proposed project. The lack of usable hydro resources indicates that this option is far from being economically attractive. Therefore, it is not suitable to develop a hydropower plant with similar capacity around the site of the proposed project. Hence the Scenario 3) is not feasible as an alternative scenario.

Step 2. Investment analysis

The purpose of this step is to determine whether the proposed project activity is economically or financially less attractive than other alternatives without an additional funding that may be derived from the CDM project activities. The investment analysis was conducted in the following steps:

Sub-step 2a. Determine appropriate analysis method

The three analysis methods suggested by *Tools for the demonstration and assessment of additionality* are simple cost analysis (option I), investment comparison analysis (option II) and benchmark analysis (option III). Since the proposed project will earn revenues from not only the CDM but also the electricity output, the simple cost analysis method is not appropriate. Investment comparative analysis method is only applicable to the case that alternative baseline scenario is similar to the proposed projects, so that comparative analysis can be conducted. The alternative baseline scenario of the proposed project is the China Southern Power Grid rather than a new investment project. Therefore option II is not an appropriate method either. The proposed project will use benchmark analysis method based on Project IRR.

Sub-step 2b . Apply benchmark analysis (Option III)

² Notice on Strictly Prohibiting the Installation of Fuel-fired Generation with the Capacity of 135MW or below issued by the General Office of the State Council, decree no. 2002-6

³ Interim Rules on the Installation and Management of Small-scale Fuel-fired Generators issued in August 1997

⁴ Source: Page 36 of *Feasibility Study Report of the proposed project*

⁵ Source: <http://www.gdwater.gov.cn/cms/xinwenjianbao/yueshuiyaowen/20051109214315218.html>



With reference to *Interim Rules on Economic Assessment of Electric Engineering Retrofit Projects*, the financial benchmark IRR of Chinese power industry is 8% the total investment, which has been used widely for Feasibility Studies of the power project investments, including wind power projects in China.

Sub-step 2c. Calculation and comparison of financial indicators

Based on the above-mentioned benchmark, the calculation and comparative analysis of financial indicators for the proposed project are carried out in sub-step 2c.

(1) Basic parameters for calculation of financial indicators

Based on the Feasibility Study Report of the proposed project, basic parameters for calculation of financial indicators are as follows:

Installed capacity: 26MW (Source: Feasibility Study Report)

Annual output: 63169MWh (Source: Feasibility Study Report)

Project lifetime: 21years (Source: Feasibility Study)

Total investment: RMB 206.719million yuan (equity/debt ratio: 33/67, source: Feasibility Study Report)

Tariff: RMB 0.528yuan/KWh (including VAT)⁶

Depreciable life: 15 years

Tax: 8.5% (VAT), 15% (income tax) (Source: Feasibility Study Report)

Expected CERs price: Euro 8.5/t CO₂e (Source: Project developer)

(2) Comparison of IRR for the proposed project and the financial benchmark

In accordance with the benchmark analysis (Option III), the proposed project will not be considered as financially attractive if its financial indicators (such as IRR) are lower than the benchmark rate.

Table 1 shows the fluctuating situation of IRR of the proposed project, with and without CDM revenues. Without the CDM revenue, the Project IRR is lower than the benchmark rate 8%. Thus the proposed project does not look financially attractive to the investors. However, with the CDM revenue, the Project IRR is significantly improved and exceeds the benchmark rate. Therefore, the proposed project with the CDM revenue can be considered as financially viable to the investors.

Table 1. Financial indicators of the Guangdong Nan Ao 26MW Wind Power Project

	Project IRR (benchmark=8%)
Without CDM revenue	6.42%
With CDM revenue	9.03%

The project IRR of the proposed project is based on the data from the Feasibility Study of the proposed project. According to the actual situation of the proposed project, as three wind turbines are not installed, the Guangdong Provincial Investigation, Design and Research Institute of Water Conservancy and Electric Power, PRC, who developed the Feasibility Study of the proposed project reevaluated the financial parameters of the proposed project and issued *Audit Opinion upon Financial Analysis of Guangdong Nan Ao 26MW Wind Power Project* on 10th November 2006⁷. With the updated financial data, the calculated project IRR is only 5.7%, which is still lower than the benchmark and much further low than the Project IRR without CDM in Table 1. It further confirmed the proposed project is financially unattractive. The Project Participant makes investment analysis with higher IRR as a conservative manner.

⁶ Source: Document No.[2004]110, Guangdong Provincial Pricing Bureau

⁷ This report is provided to the DOE.

**Sub-step 2d. Sensitivity analysis (only applicable to options II and III):**

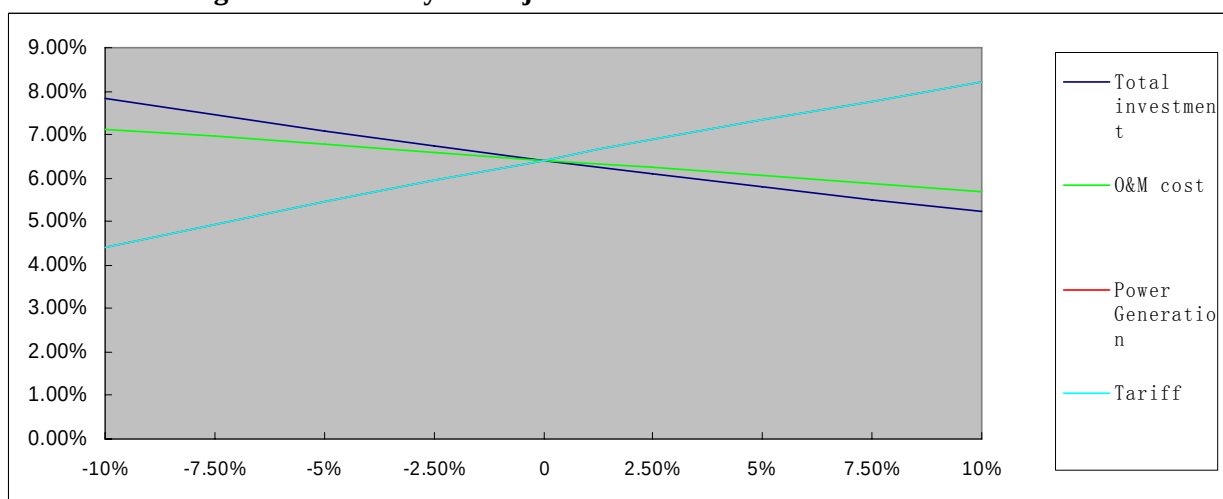
The purpose of the sensitivity analysis is to examine whether the conclusion regarding the financial viability of the proposed project is sound and tenable with those reasonable variations in the assumptions. The investment analysis provides a valid argument in favour of additionality only if it consistently supports (for a realistic range of assumptions) the conclusion that the project activity is unlikely to be the most financially attractive or is unlikely to be financially attractive.

Four financial parameters including: total investment, power generation, annual O&M cost and tariff were identified as the main variable factors for sensitive analysis of financial attractiveness. Because total investment, power generation and tariff were considered as main financial parameters for sensitive analysis and fluctuated within the range from -10% to +10% in Feasibility Study which was approved by Guangdong Provincial Development and Reform Commission. And according to the conservative principle, annual O&M cost was also considered as one of the main parameters for sensitive analysis in PDD. Their impacts on project IRR were analyzed in this step.

For detailed results of sensitive analysis of the three indicators, please see Table 2.

Table 2. Sensitivity of Project IRR to different financial parameters

Range Parameters	-10%	-7.50%	-5%	-2.50%	0	2.50%	5%	7.50%	10%
Total investment	7.82%	7.45%	7.09%	6.75%	6.42%	6.11%	5.80%	5.51%	5.23%
O&M cost	7.13%	6.96%	6.78%	6.60%	6.42%	6.24%	6.05%	5.87%	5.68%
Power generation	4.40%	4.93%	5.45%	5.94%	6.42%	6.89%	7.33%	7.77%	8.20%
Tariff	4.40%	4.93%	5.45%	5.94%	6.42%	6.89%	7.33%	7.77%	8.20%

Figure 2. Sensitivity of Project IRR to Different Financial Parameters



As shown in Table 2 and Figure 2, the Project IRR of the proposed project varies to different extents, when the above four financial indicators fluctuated within the reasonable range from -10% to +10%. In comparison, the impact of the power generation and tariff on IRR is most significant. When the power generation increases by 10%, project IRR is 8.20%, and exceeds the benchmark of 8%. The power generation in the Feasibility Study Report is calculated based on the monitored data of the local wind resources⁸ and formula that is internationally widely used⁹, which was evaluated and approved by Guangdong Development and Plan Commission. The calculated power generation is the average value over the project life time, and it has taken into account for the yearly variation in power generation, therefore, the probability of power generation increasing by 10% is very small. Furthermore, the power generation of this IRR was based on 26MW installation. Due to the fact that three wind turbines are not installed, only 23.25MW will be put into delivery, which is 10% decreased. The power generation of the proposed project will be decreased but not increased. Hence, the power generation of the proposed project is not possible to increase 10%. When the tariff of the proposed project increases by 10%, project IRR is 8.20%, and exceeds the benchmark of 8%. Before 2004, the wind farms in Guangdong Province enjoy higher tariff, which were approved case by case and over 0.62 RMB/kWh to encourage the early wind farm development¹⁰. However, to rule tariff of wind farms and reduce high cost for the power grid, Guangdong Provincial Pricing Bureau issued the tariff of 0.528RMB/kWh for new wind farms in Guangdong province¹¹. This tariff policy for wind farm in Guangdong province has not been changed since 2004. As a new developed wind farm in Guangdong province, the proposed project will be same as other new wind farm projects under the provincial tariff rule. Hence, it is not possible for the proposed project to increase tariff.

It also displays in the graphics that when the total investment and annual O&M cost varies from -10% to 10%, the project IRR is always under 8%, showing that the financial indicators have insignificant impact on IRR of the project. Considering price of raw materials has been continuously increasing in China in these years after the Feasibility Report was developed, the total investment and O&M cost will be increased, but not decreased.

The above sensitive analysis demonstrates that when financial indicators change within reasonable range, the proposed project is not financially feasible without CDM support. Hence, the Scenario 1) is not feasible as an alternative scenario.

Step 4. Common practice analysis

Sub-step 4a. Analyze other activities similar to the proposed project activity:

This common practice analysis covers similar projects within the project boundary of the proposed project, which is the area of Southern China Power Grid including Guangdong, Guangxi Zhuang Autonomous

⁸ The wind resources assessment of NanAo wind farm were assessed by 3 years wind resource assessment of the wind farm and 19 years wind resource data from NanAo weather station, Source: Chapter 2 Wind Resource, Feasibility Study of the proposed project.

⁹ WASP and WINPRO. Source: Chapter 6, Wind Turbine Selection and Location, Feasibility Study of the proposed project.

¹⁰ Source: *Develop the Wind Resource and Promote the Construction of Wind Farm* by Chen Xuefeng, Director of Research Department of Gansu Provincial Electric Power Association

¹¹ Source: Document No.[2004]110, Guangdong Provincial Pricing Bureau



Region, Guizhou and Yunnan. In April 2002, China implemented the policy "Separate power plants from network and compete in price to enter network"¹⁸. The objective of this power sector reform is to establish a more commercialized power market in China. Power project investment has to be under a more commercialized condition and considers project investment return more seriously. Since market condition for wind power project development changes much since 2002, this common practice analysis starts from 2002. Table 3 shows the similar projects (installed capacity more than 15MW^{*}) within the Southern China Power Grid since 2002, where the proposed project happens.

Table 3.a Wind Farm Development in Southern China Power Grid¹²

Project Name	Commissioned in	Total capacity	Bus-bar Tariff(Yuan/KWh)
Guangdong Shanwei Honghaiwan Wind Farm Project	2003	16.5MW	0.743 ¹³

As for Table 3.a, the Bus-bar tariff for Guangdong Shanwei Honghaiwan Wind Farm Project is much higher than that of the proposed project. The current bus-bar tariff for the newly built wind power projects is capped to a price of 0.528RMB/kWh (VAT included) according to document released by Guangdong Provincial Pricing Bureau. Such a price will also be applied to the proposed project. So the price adopted by the Project will be lower than that of Guangdong Shanwei Honghaiwan Wind Farm Project.

Sub-step 4b. Discuss any similar options that are occurring:

Table 3.b Wind Farm Development in Southern China Power Grid

Project Name	Construction Started	Total capacity	CDM activities
Shibeishan Wind Power Generation Project in Huilai County, Guangdong Province	2006.11	100.2MW	Request for registration ¹⁴
Lufeng Jiadong First Phase Wind Farm Project	2006.10	30.6MW	Registered ¹⁵

As it is demonstrated in Table 3.b, the Shibeishan Wind Power Generation Project in Huilai County, Guangdong Province is requesting for being registered as CDM project and the Lufeng Jiadong First Phase Wind Farm Project have been successfully registered as a CDM project in EB. CDM-related income is an effective measure for the project developers to overcome the barriers they encounter during project development. Hence, the development of this type of project activity in Guangdong Province is hard to be brought forth without CDM support.

¹⁸ Source: http://www.ndrc.gov.cn/xwfb/t20050708_28096.htm

* Wind farm within 15MW is defined as small scale CDM project in CDM guidance, which means wind farm project 15MW or smaller is considered to be very small comparing with big project. Hence, the bottom range for comparison with the proposed project is defined to 15MW.

¹² Source: *Statistic of installed capacity of China's Wind Farms in 2006* by Mr Shi Pengfei, Vice Chairman of Chinese Wind Energy Association Reference: <http://www.nwec.cn/Article/ShowArticle.asp?ArticleID=1131>

¹³ Source: *Develop the Wind Resource and Promote the Construction of Wind Farm* by Chen Xuefeng, Director of Research Department of Gansu Provincial Electric Power Association

¹⁴ Source: http://www.dnv.com/focus/climate_change/projects/projectdetails.asp?ProjectId=768

¹⁵ Source: <http://cdm.unfccc.int/Projects/DB/BVQ11189424843.32/view>

*Consideration of CDM before construction*

Shantou Dannan Wind Energy Co., Ltd was the former project developer of the proposed project, the biggest shareholder of which is Nuon in Netherlands. The Guangdong Development and Reform Commission approved the proposed project in Nov 2003. However, when the notice of wind tariff for Guangdong Province was issued by Guangdong Provincial Pricing Bureau in 2004, the Shantou Dannan Wind Energy Co., Ltd found the IRR of the proposed project is lower than the benchmark of 8%, which means the proposed project is not financially attractive. Shantou Dannan Wind Energy Co., Ltd was trying to find ways to improve the financial return of the project. However, being a none Chinese holding project developer, Shantou Dannan Wind Energy Co., Ltd was not eligible to develop CDM and gain CDM revenue under *Measures for Operation and Management of Clean Development Mechanism Projects in China* issued in 2005. Failing to identify approach to improve the unattractive investment return of the proposed project, Shantou Dannan Wind Energy Co., Ltd decided to quit the proposed project. And the proposed project was finally transferred it to China Resources Wind Power (Shantou) Co. Ltd who is a Chinese holding company. Guangdong Development and Reform Commission issued permission to the project developer change of the proposed project on Oct 23, 2006¹⁶ which approved the project developer change and suggested the proposed project to be developed into CDM project. The development procedure shows that the proposed project will not be developed without CDM support.

B.6. Emission reductions:**B.6.1.Explanation of methodological choices:**

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The GHG emission calculation of the proposed project was based on the instruction of ACM0002. All the data employed in the calculation is based on the available data from China Southern Power Grid. The baseline emission factor (EF_y) is calculated as a combined margin (CM), consisting of the combination of operating margin (OM) and build margin (BM) factors according to the following three steps(OM and BM of the proposed project are ex-ante):

STEP 1. Calculate the Operating Margin emission factor(s) (EF_{OM,y})

Calculation of the Operating Margin should be based on one of the four following methods according to the instruction of ACM0002:

- (a) Simple OM, or
- (b) Simple adjusted OM, or
- (c) Dispatch Data Analysis OM, or
- (d) Average OM.

Although Dispatch Data Analysis should be considered the first methodology choice as required in the ACM0002, unavailability of detailed information in China, such as the dispatch data make method (c) not feasible for the calculation in China.

¹⁶ Source: Document No. [2006]1891 Guangdong Province Development and Reform Committee



In China, specific data from the grid or each power plant is treated as business confidential and thus not publicly available. Therefore, the Simple adjusted OM (b) cannot be possibly used for the proposed project either.

Without any nuclear source, the China Southern Power Grid only possesses 30.938% of its total electricity generation that come from renewable energy sources in 2005, 29.912% in 2004, 41.752% in 2003, 32.983% in 2002, 33.705% in 2001¹⁷. Hence, the low operating cost/must run sources is much less than 50% of the total grid generation, which accords with the defined condition of method (a), but not method (d). Consequently, Simple OM method is selected to calculate the Operating Margin emission factor of the proposed project.

The Simple OM emission factor ($EF_{OM,simple,y}$) is calculated as the generation-weighted average emissions per electricity unit (t CO₂e/MWh) of all generating sources serving in the system, excluding low-operating cost and must-run power plants:

$$EF_{OM,simple,y} = \frac{\sum_{i,j} F_{i,j,y} \cdot COEF_{i,j,y}}{\sum_j GEN_{j,y}}$$

Where:

- $F_{i,j,y}$ the amount of fuel i consumed by relevant power sources j in year(s) y . The index j runs over all power sources including imports, but excludes low operating costs and must-run power plants.
- $COEF_{i,j,y}$ the CO₂ emission coefficient of fuel i , taking into account the carbon content of fuels used by relevant power sources j and the percentage oxidation of the fuel in year(s) y ;
- $GEN_{j,y}$ the electricity delivered to the grid by source j in year y .

The CO₂ emission coefficient is equal to the net calorific value of fuel i , multiplied by the oxidation factor of the fuel and the CO₂ emission factor per unit of energy of the fuel i .

$$COEF_i = NCV_i \cdot EF_{CO2,i} \cdot OXID_i$$

NCV_i is the net calorific value (TJ per mass or volume unit) of fuel i ;

$OXID_i$ is the oxidation factor of the fuel. IPCC default values are used.

$EF_{CO2,i}$ is the CO₂ emission factor per unit of energy of the fuel i (tCO₂e/TJ). $EF_{CO2,i}$ of fossil fuels is from IPCC defaults.

STEP 2. Calculate the Build Margin emission factor ($EF_{BM,y}$)

The Build Margin Emission Factor is calculated as the generation weighted average emission factor (measured in tCO₂e/MWh) of a sample of m power plants:

$$EF_{BM,y} = \frac{\sum_{i,m} F_{i,m,y} \times COEF_{i,m,y}}{\sum_m GEN_{m,y}}$$

¹⁷ China Electric Power Yearbook



Where

- $F_{i,m,y}$ the amount of fuel i consumed by relevant power sources m in year(s) y . The index m runs over all power sources including imports.
- $COEF_{i,m,y}$ the CO₂ emission coefficient of fuel i , taking into account the carbon content of fuels used by relevant power sources m and the percentage oxidation of the fuel in year(s) y ;
- $GEN_{m,y}$ the electricity delivered to the grid by source m in year y .

ACM0002 provides two options for calculating $EF_{BM,y}$:

Both options have the same requirement on sample group m : either the five power plants built most recently, or the power plants capacity additions in the electricity system that comprising 20% of the system generation (in MWh) and that have been built most recently. The project chooses ex-ante $EF_{BM,y}$

However, it is very difficult to obtain the data of the five power plants built most recently because these data are considered as confidential information by the company itself and the Grid in China. Therefore, a deviation approved by the EB is applied here in the calculation that is to calculate the new capacity additions and the proportion of each technology of power generation. Then the weighing of capacity additions of different technologies will be worked out. Finally the emission factor will be calculated by employing the efficiency factor representing the best technology commercially available.

Deviated Calculation of Build Margin (BM):

Sub-step 1. Calculation of weights of CO₂ emissions of solid, liquid and gas fuel in total emissions for power generation

$$\lambda_{Coal} = \frac{\sum_{i \in COAL, j} F_{i,j,y} \times COEF_{i,j}}{\sum_{i,j} F_{i,j,y} \times COEF_{i,j}}$$

$$\lambda_{Oil} = \frac{\sum_{i \in OIL, j} F_{i,j,y} \times COEF_{i,j}}{\sum_{i,j} F_{i,j,y} \times COEF_{i,j}}$$

$$\lambda_{Gas} = \frac{\sum_{i \in GAS, j} F_{i,j,y} \times COEF_{i,j}}{\sum_{i,j} F_{i,j,y} \times COEF_{i,j}}$$

Where:

$F_{i,j,y}$: the consumption of fuel i for province j in year y (tce);

$COEF_{i,j,y}$: the emission factor (tCO₂/tce) of fuel i , taking into account the carbon content of fuel i and the percentage of oxidation of the fuel in year y ;

COAL, OIL and GAS respectively refers to the group of solid, liquid, and gas fuels.



Sub-step 2: Calculation of Emission Factor of Relevant Thermal Power

$$EF_{Thermal} = \lambda_{Coal} \times EF_{Coal, Adv} + \lambda_{Oil} \times EF_{Oil, Adv} + \lambda_{Gas} \times EF_{Gas, Adv}$$

Where: $EF_{Coal, Adv}$, $EF_{Oil, Adv}$ and $EF_{Gas, Adv}$ respectively refers to the emission factor representing best technology commercially available for fuel of coal, oil or gas fired power plants.

Sub-step 3: Calculation of BM of the Grid

$$EF_{BM, y} = \frac{CAP_{Thermal}}{CAP_{Total}} \times EF_{Thermal}$$

Where:

CAP_{Total} is the total of new capacity additions;

$CAP_{Thermal}$ is the new capacity addition of thermal power.

STEP 3. Calculate the baseline emission factor EF_y

The Baseline Emission Factor is calculated as a Combined Margin, using a weighted average of the Operating Margin and Build Margin.

The default values of weighted factor are:

$$w_{OM} = 0.75 \quad w_{BM} = 0.25$$

$$EF_y = EF_{OM} * w_{OM} + EF_{BM} * w_{BM}$$

Baseline Emissions are calculated by multiplying the ex-ante Baseline Emission factor by annual power supply to the Grid.

$$BE_y = EG_y * EF_y$$

With:

BE_y the baseline emission of China Southern Power Grid in year y

EG_y the amount of power generated by the project and supplied to the grid

EF_y the ex-ante emission factor in year y

$$EG_y = EG_{output, y} - EG_{import, y}$$

With:

EG_y the net Electricity supplied to the grid by the project in year y

$EG_{output, y}$ the Electricity supplied to the grid by the project in year y

$EG_{import, y}$ the Electricity purchased from the grid by the project in year y

Emissions from project activity

The Proposed Project is a zero-emission electricity generating activity; therefore no emissions from the project activity were identified.

**Leakage**

The project does not consider leakage according to the requirements of methodology applied.

Emission Reductions

The annual emission reductions ER_y for the project activity are calculated as the baseline emissions minus the project emissions. Being a project of zero-emission activity the final GHG emission reductions are calculated as follows:

$$ER_y (tCO_2e/yr) = E_{By} - E_{Py} = (EG_y * EF_y) - 0$$

Where:

$EG_y (MWh/yr)$ = Electricity supplied to the Grid by the project each year;

$EF_y (tCO_2e/MWh)$ = GHG emission factor of the China Southern Power Grid (ex-ante).

B.6.2. Data and parameters that are available at validation:

Data / Parameter:	EF_{OM}
Data unit:	tCO ₂ / MWh
Description:	Operating Margin Emission Factor
Source of data used:	Baseline Emission Factors for Power Grids in China, sourced from China Energy Statistical Yearbook, China Electric Power Yearbook and 2006 IPCC Guidelines for National Greenhouse Gas Inventories
Value applied:	1.0119
Justification of the choice of data or description of measurement methods and procedures actually applied :	Calculated according to the updated Baseline Emission Factors for Power Grids in China based on ACM0002 and EB guidance
Any comment:	

Data / Parameter:	EF_{BM}
Data unit:	tCO ₂ / MWh
Description:	Build Margin Emission Factor
Source of data used:	Baseline Emission Factors for Power Grids in China, sourced from China Energy Statistical Yearbook, China Electric Power Yearbook and 2006 IPCC Guidelines for National Greenhouse Gas Inventories
Value applied:	0.6748
Justification of the choice of data or description of measurement methods and procedures actually applied :	Calculated according to the updated Baseline Emission Factors for Power Grids in China based on ACM0002 and EB guidance



Any comment:	
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Data / Parameter:	EF_y
Data unit:	tCO ₂ / MWh
Description:	Combined Emission Factor
Source of data used:	Baseline Emission Factors for Power Grids in China, sourced from China Energy Statistical Yearbook, China Electric Power Yearbook and 2006 IPCC Guidelines for National Greenhouse Gas Inventories
Value applied:	0.9276
Justification of the choice of data or description of measurement methods and procedures actually applied :	Calculated according to the updated Baseline Emission Factors for Power Grids in China based on ACM0002 and EB guidance
Any comment:	

Data / Parameter:	F_{i,j,y}
Data unit:	tonne, m ³
Description:	the amount of fuel <i>i</i> (in a mass or volume unit) consumed by relevant power sources <i>j</i> in year(s) <i>y</i>
Source of data used:	China Energy Statistical Yearbook (2003~2006)
Value applied:	See Annex 3 for details
Justification of the choice of data or description of measurement methods and procedures actually applied :	Official statistical data
Any comment:	

Data / Parameter:	NCV_i
Data unit:	MJ / tonne or km ³
Description:	Net Calorific Value of fossil fuel type <i>I</i> consumed by the power plants in the grid
Source of data used:	China Energy Statistical Yearbook (2006)
Value applied:	See Annex 3 for details
Justification of the choice of data or description of measurement methods and procedures actually applied :	Official statistical data
Any comment:	

Data / Parameter:	EF_{CO2,i}
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Data unit:	tc/TJ
Description:	CO2 emission factor of fossil fuel type <i>I</i> consumed by the power plants in the grid
Source of data used:	2006 IPCC Guidelines for National Greenhouse Gas Inventories
Value applied:	See Annex 3 for details
Justification of the choice of data or description of measurement methods and procedures actually applied :	IPCC default value
Any comment:	

Data / Parameter:	Carbon Oxidation Factor
Data unit:	%
Description:	Carbon Oxidation Factor of fossil fuel type <i>I</i> consumed by the power plants in the grid
Source of data used:	2006 IPCC Guidelines for National Greenhouse Gas Inventories
Value applied:	See Annex 3 for details
Justification of the choice of data or description of measurement methods and procedures actually applied :	IPCC default value
Any comment:	

Data / Parameter:	Installed Capacity
Data unit:	MW
Description:	The Installed Capacity of the power plants in the grid in the year <i>y</i>
Source of data used:	China Electric Power Yearbook(2004-2006)
Value applied:	See Annex 3 for details
Justification of the choice of data or description of measurement methods and procedures actually applied :	Official statistical data
Any comment:	

Data / Parameter:	Electricity Generation
Data unit:	MWh
Description:	The electricity generation of the power plants in the grid in the year <i>y</i>
Source of data used:	China Electric Power Yearbook(2004-2006)
Value applied:	See Annex 3 for details
Justification of the	Official statistical data



choice of data or description of measurement methods and procedures actually applied :	
Any comment:	

Data / Parameter:	Electricity self-consumption ratio
Data unit:	%
Description:	The ratio of electricity self-consumption to the total electricity generation of the power plants
Source of data used:	China Electric Power Yearbook(2004-2006)
Value applied:	See Annex 3 for details
Justification of the choice of data or description of measurement methods and procedures actually applied :	Official statistical data
Any comment:	

B.6.3 Ex-ante calculation of emission reductions:
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>>

As described in B.6, the emission reductions of the proposed project are calculated as follows:

Baseline emissions

Annual generation (net of auxiliary power i.e. the on-site electricity usage for the operation of the wind farm) is estimated as 57,595 MWh.

The ex-ante baseline emission factor: 0.9276 tCO₂/ MWh

Annual baseline emissions: 53,425 tCO₂ (details in Annex 3)

Project emissions

According to the baseline methodology ACM0002, the GHG emission of the proposed project within the project boundary is zero, i.e.

$$E_{Py}=0$$

Leakage

According to the baseline methodology ACM0002, the leakage of the proposed project is not considered,

$$L_{Py} = 0$$

**Project Emission Reductions**

Net emission reductions of the proposed project = Total baseline emissions – Total project emissions

The total annual baseline emissions are **53,425 tCO₂**

The total annual project emissions are **0 tCO₂**.

The annual emission reductions are estimated to be: **53,425 tCO₂**. The proposed project activity is expected to achieve **373,975 tCO₂e** of net emission reductions during the first 7-year crediting period.

B.6.4 Summary of the ex-ante estimation of emission reductions:

>>

Year	Estimation of project activity emissions (tonnes of CO ₂ e)	Estimation of baseline emissions (tonnes of CO ₂ e)	Estimation of leakage (tonnes of CO ₂ e)	Estimation of overall emission reductions (tonnes of CO ₂ e)
2008(Nov.-Dec.)	0	8,904	0	8,904
2009	0	53,425	0	53,425
2010	0	53,425	0	53,425
2011	0	53,425	0	53,425
2012	0	53,425	0	53,425
2013	0	53,425	0	53,425
2014	0	53,425	0	53,425
2015(Jan.-Oct.)	0	44,521	0	44,521
Total (tonnes of CO₂e)	0	373,975	0	373,975

B.7 Application of the monitoring methodology and description of the monitoring plan:**B.7.1 Data and parameters monitored:**

(Copy this table for each data and parameter)

Data / Parameter:	EG _{output}
Data unit:	MWh
Description:	Electricity generated by the proposed project during year y
Source of data to be used:	Electricity meter reading at the substation between the proposed project and the Grid
Value of data applied for the purpose of calculating expected emission reductions in section B.5	59,376
Description of measurement methods	The readings of the electricity meter will be hourly measured and monthly recorded. Data will be archived for 2 years following the end of the crediting



and procedures to be applied:	period by means of electronic and paper backup. The accuracy of the meter is 0.5s. The national Calibration standard (dl/t 448-2000) will be applied in the project. The calibration frequency is one time/year.
QA/QC procedures to be applied:	The electricity generation from the plant will be monitored and recorded by the electricity meter. The project operator is responsible for recording this set of data. Receipts from electricity sales will also be obtained for double check.
Any comment:	

Data / Parameter:	EG _{import,y}
Data unit:	MWh
Description:	Electricity purchased from the Grid by the proposed project during year y
Source of data to be used:	Electricity meter reading at the substation between the proposed project and the Grid
Value of data applied for the purpose of calculating expected emission reductions in section B.5	1,781
Description of measurement methods and procedures to be applied:	The readings of the electricity meter will be hourly measured and monthly recorded. Data will be archived for 2 years following the end of the crediting period by means of electronic and paper backup. The accuracy of electricity meter is 0.5s. The national Calibration standard (dl/t 448-2000) will be applied in the project. The calibration frequency is one time/year.
QA/QC procedures to be applied:	The electricity purchased by the project will be monitored and recorded by the electricity meter. The project operator is responsible for recording such data. Receipts for electricity sales will be kept for further verification, when necessary.
Any comment:	

B.7.2 Description of the monitoring plan:

>>

>

This Monitoring plan will set out a number of monitoring tasks in order to ensure that all aspects of projected greenhouse gas (GHG) emission reductions for the proposed project are controlled and reported. This requires an on going monitoring of the project to ensure performance according to its design and that claimed Certified Emission Reductions (CERs) are actually achieved.

The monitoring plan provides the requirements and instructions for:

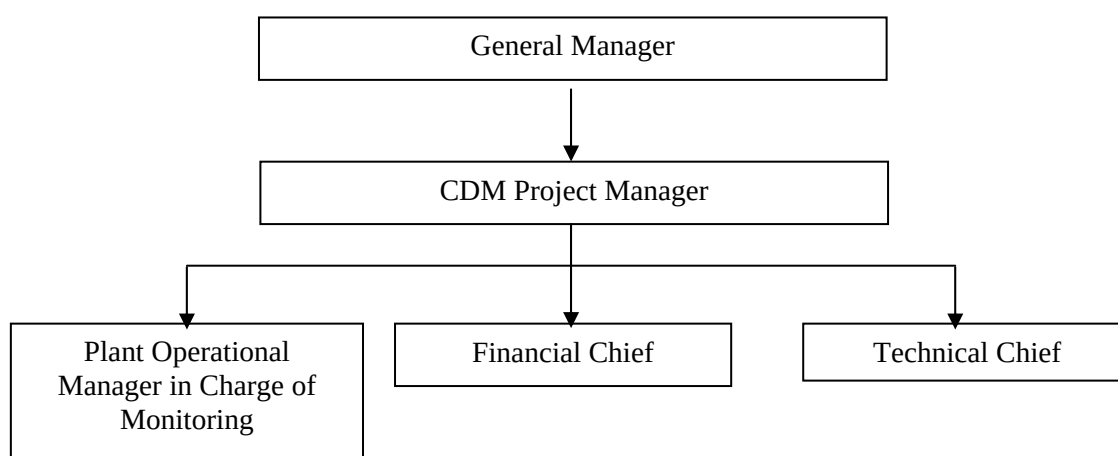
- ℓ Establishing and maintaining the appropriate monitoring systems for electricity generated by the project;
- ℓ Quality control of the measurements;
- ℓ Procedures for the periodic calculation of GHG emission reductions;
- ℓ Assigning monitoring responsibilities to personnel;
- ℓ Data storage and filing system;
- ℓ Preparing for the requirements of an independent, third party auditor or verifier.

1. User of the Monitoring Plan



China Resources Wind Power (Shantou) Co., Ltd, the proposed project owner, will use this document as guideline in monitoring of the project emission reduction performance and will adhere to the guidelines set out in this monitoring plan. This plan should be modified according to actual conditions and requirements of DOE in order to ensure that the monitoring is credible, transparent, and conservative.

The responsibilities for carrying out these tasks are illustrated in the table below:



Operational manager of wind farm will collect the information and data required by the Monitoring Plan. The collected information will be documented and sent to the CDM manager and responsible staffs of the China Resources Wind Power (Shantou) Co., Ltd. monthly. The CDM manager will in charge of the implementation of the Monitoring Plan and report to the General Manager of the company. The General Manager of the company will make the confirmations on monitoring, calculation data and reports.

2. Calibration of Meters & Metering

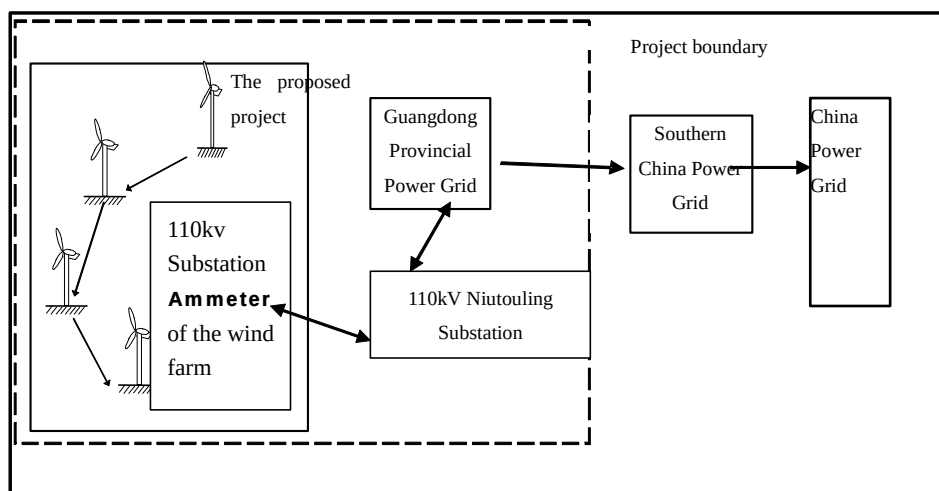
The accuracy of electricity meter is 0.5s that meets the national requirement and the calibration frequency is one time/year. The national Calibration standard (dl/t 448-2000) will be applied in the proposed project. An agreement should be signed between the proposed project owner and the Grid that defines the metering arrangements and the required quality control procedures to ensure accuracy. The metering equipment will be properly calibrated and checked annually for accuracy. The project owner will prepare backup procedures to deal with any errors occurred to the meters. In case of any errors happens, the grid-connected electricity generated by the proposed project shall be determined by the project owner and the Grid jointly according to the error handling procedures.

The calibration records is carried out by the grid company with the records being provided to the proposed project owner, and these records will be maintained by the proposed project owner and the third party designated.

3. Monitoring

Grid-connected electricity generated by the proposed project will be monitored through metering equipment (ammeter monitoring both electricity input and electricity output) at the substation (interconnection facility connecting the facility to the grid) which locates in the project site. The ammeter will be used to monitor the grid-connected electricity generated by the proposed project generally but

when the wind farm is not in operation and electricity supplied from the Grid will be used to ensure the minimum requirement of running the wind farm, the ammeter can also be used to monitor the power purchased from the Grid. The data can also be monitored and recorded at the on-site control centre using a computer system. The meter reading will be readily accessible for DOE. Calibration tests records will be maintained for verification.



4. Quality Assurance and Quality Control

The quality assurance and quality control procedures for recording, maintaining and archiving data shall be improved as part of this CDM project activity. This is an on-going process that will be ensured through the CDM in terms of the need for verification of the emissions on an annual basis according to this PDD and the CDM manual.

5. Data Management System

This provides information on record keeping of the data collected during monitoring. Record keeping is the most important exercise in relation to the monitoring process. Without accurate and efficient record keeping, project emission reductions cannot be verified. Below follows an outline of how project related records would be managed.

Overall responsibility for monitoring of GHG emissions reduction will rest with the CDM responsible person of the proposed project. The CDM manual sets out the procedures for tracking information from the primary source to the end-data calculations in paper document format. It is the responsibility of the proposed project owner to provide additional necessary data and information for validation and verification requirements of respective DOE. Physical documentation such as paper-based maps, diagrams and environmental assessment will be collated in a central place, together with this monitoring plan. All paper-based information will be stored by the proposed project owner and kept at least one copy.

Table List of the key documents relevant to monitoring and verification

No.	Document Title	Main Content	Source
1	Project Design Document (PDD)	Calculation procedure of emission reduction and monitoring items	Proposed project owner or UNFCCC website
2	Monitoring Quality Control and Quality Assurance Report	Equipments and national and industry standards	Proposed project owner



3	The report on qualifications of the persons responsible for the monitoring and calculation	For example, the title of a technical post, working experience etc.	Proposed project owner
4	The report on monitoring and checking of electricity supplied to the grid	Record based on monthly meter reading and electricity sale receipts	Proposed project owner
5	Record on maintenance and calibration of metering equipment	Reasons for maintenance and calibration and the precision after maintenance and calibration	Proposed project owner
6	The report on baseline emission factor calculation	Data sources and calculation procedure based on the information in the PDD	Proposed project owner
7	Record on CO2 emission reduction	Monthly calculation based on the formula in the PDD	Proposed project owner
8	Letter of confirmation on No. 2 to No.7	Make confirm of monitoring and calculation data and procedure from No. 2 to No.7	Proposed project owner
9	Project Management Record (including data collection and management system)	Comprehensively and truly reflect the management and the operation of the proposed project	Proposed project owner

6. Verification and Monitoring Results

The verification of the monitoring results of the project is a mandatory process required for all CDM projects. The main objective of the verification is to independently verify that the project has achieved the emission reductions as reported and projected in the PDD. It is expected that the verification will be done annually.

B.8 Date of completion of the application of the baseline study and monitoring methodology and the name of the responsible person(s)/entity(ies)

>>

Date of completion of Baseline Study: 19/10/2007

Name of person/entity determining the baseline:

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**SECTION C. Duration of the project activity / crediting period****C.1 Duration of the project activity:****C.1.1. Starting date of the project activity:**

>>

30/04/2007

C.1.2. Expected operational lifetime of the project activity:

>>

21 years

C.2 Choice of the crediting period and related information:**C.2.1. Renewable crediting period****C.2.1.1. Starting date of the first crediting period:**

>>

01/11/2008 (or the date after registration whichever is later)

C.2.1.2. Length of the first crediting period:

>>

7 years

C.2.2. Fixed crediting period:**C.2.2.1. Starting date:**

>>

Not applicable

C.2.2.2. Length:

>>

Not applicable

**SECTION D. Environmental impacts**

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D.1. Documentation on the analysis of the environmental impacts, including transboundary impacts:

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The environmental impact assessment for this project was carried out by Shantou Environmental Protection Research Institute in July 2004. And approved by Shantou Environmental Protection Bureau on August 13th 2004. A summary of the report is illustrated as below:

Impact on natural environment

The lands permanently taken by wind farms are normally mountain ridges, where most of the vegetation is grass and shrub, without rare plants. Hence, the project construction has little impact on the mountain's eco-environment. The proposed wind power project will not ruin the local scenery. Additionally, it will become a new prospect.

No migrating birds have been found in the project field till now. Therefore, the project is not located on the passage of migrating birds, and the project construction will not influence the migration of birds.

Impact from noise

There is some noise during the operation of wind turbines. However, no village or resident exists around the project location. Consequently, the noise of operation has little impact to the surrounding environment. Besides, the noise level of the advanced wind turbines employed in the proposed project accords with the National Standard (Grade II of *the Noise Standard for City Environment GB3096-93*: daytime 60dB). As a result, the noise will not impact the work and daily life of local residents.

Electromagnetic impact

The operation of the wind farm will generate electromagnetic pollution, whereas the pollution is slight. In addition, the project is located on the mountains, with no resident or village around it. Therefore, the electronic magnetic pollution to the surrounding environment is insignificant.

Impacts on water quality

Wastewater is mainly domestic wastewater. Wastewater quantity is fairly small and treatment methods will be applied for on-site primary treatment, and then the wastewater will be removed periodically. Therefore, the impact of wastewater is limited and mitigated. All the wind turbines applied here are cooled by natural wind without the need of water usage. Therefore, the project will not influence the water quality.

Wind power doesn't generate direct pollution; neither does it involve any farmland occupancy or resident migration. Therefore, there is no significantly negative impact to the eco-environment after the project construction.

D.2. If environmental impacts are considered significant by the project participants or the host Party, please provide conclusions and all references to support documentation of an environmental impact assessment undertaken in accordance with the procedures as required by the host Party:

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The environmental impacts of the Guangdong Nan Ao 26MW Wind Power Project are not considered significant.

**SECTION E. Stakeholders' comments**

>>

E.1. Brief description how comments by local stakeholders have been invited and compiled:

>>

The project developer has sent out questionnaires to the stakeholders in the Nan Ao County for the comments of the proposed project construction in April 2007. 50 copies of questionnaire were distributed, and 48 pieces of reply were received. The recovery ratio is 96%. Among the interviewees, there were 6 farmers, 17 workers, 14 government official, 8 students, and 3 others. 15 of them have educational level of middle school, 11 high school, 14 college, and 8 others. The questions investigated were mainly as follows:

1. Whether the environment of your life, work and study is quiet or not?
2. Whether there is any electromagnetic wave disturbance in receiving TV signal?
3. Whether the proposed project has any adverse impact on life?
4. Whether the project will improve the living and working conditions?
5. What the impact on the sound environment quality is?
6. What is you major concern in the proposed project?
7. Whether support the project?
8. What should be pay attention in the construction and operation of the proposed project?
9. Is there any comment and suggestion to the project operator?

E.2. Summary of the comments received:

>>

The summary of survey is listed as the following:

Among the 48 interviewees, all of the investigated people thought that the environment of life, work and study was quiet; 47 people thought that there were no disturbance of the TV signal receiving, and only 1 people was not sure about that; 41 of them believed that the project had no negative impact on their life and 7 people were not sure; 43 people believed that the project will improve the living condition, 2 people were not sure about that, and just 3 people thought it will not; 9 of them thought it had no negative impact on the sound environment, 6 people thought some negative impact were existing, and 33 people had no idea; As to the environmental problems aroused during the construction and operational period of the proposed project, 42 of them conceded noise may be the most important disturbance, 3 of them concerned the disturbance of the TV signal receiving, while 3 of them worry about the waste water discharge. However, it can be learned from the survey summary that the interviewees fully support the construction of the project.

E.3. Report on how due account was taken of any comments received:

>>

The local community possesses basically strong positive comments on the effects of the proposed project. However, several interviewees were not sure or had no idea about some questions. The reason is that these interviewees are not familiar with the proposed project and related environmental protection knowledge. After the investigator made a detailed introduction to them, they expressed their positive comments and supported the project construction. During the survey some people express their concerns about the environmental impacts of the project, such as electromagnetic disturbance, noise issues during construction. These concerns have been well analysed and treated in the EIA report, and measures for mitigating the negative impacts on local bio-environment and domestic living of local people have been well planned. With good implementation of the above-mentioned measures, the proposed project will not



have significant negative impact on the local environment. There has, therefore, been no reason to modify the plans due to comments received.

**Annex 1****CONTACT INFORMATION ON PARTICIPANTS IN THE PROJECT ACTIVITY**

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Annex 2

INFORMATION REGARDING PUBLIC FUNDING

There is no public funding for the Guangdong Nan Ao 26MW Wind Power Project.

**Annex 3****BASELINE INFORMATION**

All the tables related to the calculation of baseline emission reduction are presented below:

Calculation of Operating Margin (OM):**Table A1. Simple OM Emission Factor of Southern China Power Grid in 2003**

Fuel types	Unit	Guangdong	Guangxi	Guizhou	Yunnan	Subtotal	Emission Factor (t/TJ)	Oxidation rate (%)	Average low Caloric value (MJ/t _{km} ³)	CO ₂ emission (tCO _{2e}) $I=G*H*F*E*44/12/10000$ (unit of mass)
		A	B	C	D	E=A+B+C+D	F	G	H	$I=G*H*F*E*44/12/10000$ (unit of volume)
Raw coal	10000ton	4491.79	831.84	2169.11	1405.27	8898.01	25.8	100	20908	175993455.05
Cleaned coal	10000ton	0.05				0.05	25.8	100	26344	1246.07
Other washed coal	10000ton			36.38	20.37	56.75	25.8	100	8363	448971.84
Coke	10000ton				0.5	0.5	25.8	100	28435	13449.76
Coke oven gas	10 ⁸ m ³				0.04	0.04	12.1	100	16726	2968.31
Other coal gas	10 ⁸ m ³	3.21			11.27	14.48	12.1	100	5227	335797.81
Crude oil	10000ton	6.85				6.85	20	100	41816	210055.71
Gasoline	10000ton	0.02				0.02	18.9	100	43070	596.95
Diesel	10000ton	31.9			0.76	32.66	20.2	100	42652	1031759.27

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Fuel oil	10000 ton	627.22	0.3			627.52	21.1	100	41816	20301304.48
LPG	10000 ton					0	17.2	100	50179	0.00
Refinery gas	10000 ton	2.85				2.85	18.2	100	46055	87592.00
Natural gas	10 ⁸ m ³					0	15.3	100	38931	0.00
Other oil product	10000 ton	11.35				11.35	20	100	38369	319357.98
Other coking product	10000 ton					0	25.8	100	28435	0.00
Other fuel	10000 tce	93.21			22.35	115.56	0	100	0	0.00
									I. Subtotal	198746555.23

China Energy Statistical Yearbook 2004

Table A2. Thermal Power Generation of Southern China Power Grid in 2003

Province	Power Generation (MWh)	Ratio of Self Power Consumption of Plant (%)	Power Supply (MWh)
Guangdong	143351000	5.5	135466695
Guangxi	17079000	8.43	15639240
Guizhou	43295000	7.4	40091170
Yunnan	19055000	8.01	17528695
Total			208725800

China Electric Power Yearbook 2004

**Table A3. Emission Factor of Southern China Power Grid in 2003**

	Parameter	Unit	Value	Source
A	Net Import from Central China Power Grid to South Power Grid (MWh)	MWh	11100	China Power Yearbook 2004
B	Total Emissions of Central China Power Grid	tCO ₂ e	276404544	Calculation based on data from China Power Yearbook and China Energy Statistical Yearbook
C	Total Power Supply of Central China Power Grid	MWh	225987719	China Power Yearbook 2004
D	Average Emission Factor of Central China Power Grid	tCO ₂ e/MWh	0.797442	
E	Total Power Supply of Southern China Power Grid	MWh	208736900	E=Total Power Generation of South Power Grid + A
F	Total Emissions of Southern China Power Grid	tCO ₂ e	198755407	
G	Emission Factor of Southern China Power Grid	tCO ₂ e/MWh	0.952181	G=F/E

Table A4. Simple OM Emission Factor of Southern China Power Grid in 2004

Fuel types	Unit	Guangdong	Guangxi	Guizhou	Yunnan	Subtotal	Emission Factor (tC/TJ)	Oxidation rate (%)	Average low Caloric value (MJ/t, km ³)	CO ₂ emission (tCO ₂ e) $I=G*H*F*E*44/12/100$ 00 (unit of mass)
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		A	B	C	D	E=A+B+C+D	F	G	H	I=G*H*F*E*44/12/100 00 (unit of volume)
Raw coal	10000 ton	6017.7	1305	2643.9	1751.28	11717.88	25.8	100	20908	231767573.55
Cleaned coal	10000 ton	0.21				0.21	25.8	100	26344	5233.50
Other washed coal	10000 ton					0	25.8	100	8363	0.00
Coke	10000 ton					0	25.8	100	28435	0.00
Coke oven gas	10 ⁸ m ³					0	12.1	100	16726	0.00
Other coal gas	10 ⁸ m ³	2.58				2.58	12.1	100	5227	59831.38
Crude oil	10000 ton	16.89				16.89	20	100	41816	517932.98
Gasoline	10000 ton					0	18.9	100	43070	0.00
Diesel	10000 ton	48.88			1.83	50.71	20.2	100	42652	1601975.28
Fuel oil	10000 ton	957.71				957.71	21.1	100	41816	30983494.25
LPG	10000 ton					0	17.2	100	50179	0.00
Refinery gas	10000 ton	2.86				2.86	18.2	100	46055	87899.34
Natural gas	10 ⁸ m ³	0.48				0.48	15.3	100	38931	104833.40
Other oil product	10000 ton	1.66				1.66	20	100	38369	46707.86
Other coking product	10000 ton					0	25.8	100	28435	0.00
Other fuel	10000 tce	79.42				79.42	0	100	0	0.00
									Subtotal	265175481.54

China Energy Statistical Yearbook 2005

**Table A5. Thermal Power Generation of Southern China Power Grid in 2004**

Province	Power Generation (MWh)	Ratio of Self Power Consumption of Plant (%)	Power Supply (MWh)
Guangdong	169389000	5.42	160208116
Guangxi	20143000	8.33	18465088
Guizhou	49720000	7.06	46209768
Yunnan	24322000	7.56	22483257
Total			247366229

China Electric Power Yearbook 2005

Table A6. Emission Factor of Southern China Power Grid in 2004

Parameter	Unit	Value	Source	Parameter
A	Net Import from Central China Power Grid to South Power Grid (MWh)	MWh	10951240	China Power Yearbook 2005
B	Total Emissions of Central China Power Grid	tCO ₂ e	345671697	Calculation based on data from China Power Yearbook and China Energy Statistical Yearbook
C	Total Power Supply of Central China Power Grid	MWh	249074186	China Power Yearbook 2005
D	Average Emission Factor of Central China Power Grid	tCO ₂ e/MWh	0.826448	
E	Total Power Supply of Southern China Power Grid	MWh	258317469.1	E=Total Power Generation of South Power Grid + A
F	Total Emissions of Southern China Power Grid	tCO ₂ e	274226117	



G	Emission Factor of Southern China Power Grid	tCO ₂ e/MWh	1.061586	G=F/E
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Table A7. Simple OM Emission Factor of Southern China Power Grid in 2005

Fuel types	Unit	Guangdong	Guangxi	Guizhou	Yunnan	Subtotal	Emission Factor (t/TJ)	Oxidation rate (%)	Average low Caloric value (MJ/t, km ³)	CO ₂ emission (tCO ₂ e) $I = G * H * F * E * 44 / 12 / 100$ 00 (unit of mass)
		A	B	C	D	E=A+B+C+D	F	G	H	$I = G * H * F * E * 44 / 12 / 100$ 00 (unit of volume)
Raw coal	10000ton	6696.47	1435	3212.31	1975.55	13319.33	25.8	100	20908	263442601.85
Cleaned coal	10000ton				0.15	0.15	25.8	100	26344	3738.21
Other washed coal	10000ton			10.39	33.88	44.27	25.8	100	8363	350237.59
Coke	10000ton	4.79			8.05	12.84	25.8	100	28435	345389.71
Coke oven gas	10 ⁸ m ³			3212.31	0.79	0.79	12.1	100	16726	58624.07
Other coal gas	10 ⁸ m ³	1.87			15.96	17.83	12.1	100	5227	413485.84
Crude oil	10000ton	10.91				10.91	20	100	41816	334555.88
Gasoline	10000ton	0.68				0.68	18.9	100	43070	20296.31
Diesel	10000ton	31.96	2.02		1.81	35.79	20.2	100	42652	1130638.84
Fuel oil	10000ton	887.21				887.21	21.1	100	41816	28702703.26
LPG	10000ton					0	17.2	100	50179	0.00
Refinery gas	10000ton	4.92				4.92	18.2	100	46055	151211.46
Natural gas	10 ⁸ m ³	0.93				0.93	15.3	100	38931	203114.71
Other oil product	10000ton	1.7				1.7	20	100	38369	47833.35
Other coking	10000ton					0	25.8	100	28435	0.00



product										
Other fuel	10000 tce	104.66	133.15		59.72	297.53	0	100	0	0.00
									Subtotal	295204431.07

China Energy Statistical Yearbook 2006

Table A8. Thermal Power Generation of Southern China Power Grid in 2005

Province	Power Generation (MWh)	Ratio of Self Power Consumption of Plant (%)	Power Supply (MWh)
Guangdong	176453000	5.58	166606923
Guangxi	25023000	7.95	23033672
Guizhou	58430000	7.34	54141238
Yunnan	27281000	6.94	25387699
Total			269169531
China Electric Power Yearbook 2006			

Table A9. Emission Factor of Southern China Power Grid in 2005

Parameter	Unit	Value	Source	Parameter
A	Net Import from Central China Power Grid to South Power Grid (MWh)	MWh	96363000	China Power Yearbook 2006
B	Total Emissions of Central China Power Grid	tCO ₂ e	359887488	Calculation based on data from China Power Yearbook and China Energy Statistical Yearbook



C	Total Power Supply of Central China Power Grid	MWh	286203305	China Power Yearbook 2006
D	Average Emission Factor of Central China Power Grid	tCO ₂ e/MWh	0.771225	
E	Total Power Supply of Southern China Power Grid	MWh	365532531	E=Total Power Generation of South Power Grid + A
F	Total Emissions of Southern China Power Grid	tCO ₂ e	369521975	
G	Emission Factor of Southern China Power Grid	tCO ₂ e/MWh	1.010914	G=F/E

Table A10. Operating Margin Emission Factor of Southern China Power Grid

		Year 2003	Year 2004	Year 2005	Total
A	Emissions (tCO ₂ /year)	198755407	274226117	369521975	842503499
B	Power Supply (MWh)	208736900	258317469.1	365532531	832586900
C	CO ₂ Emission Factor (tCO ₂ /MWh)	C = A/B			1.011911

Calculation of Build Margin (BM):

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Step 1. Calculation of weights of CO₂ emissions of solid, liquid and gas fuel in total emissions for power generation

$$\lambda_{Coal} = \frac{\sum_{i \in COAL, j} F_{i,j,y} \times COEF_{i,j}}{\sum_{i,j} F_{i,j,y} \times COEF_{i,j}} \quad (2)$$

$$\lambda_{Oil} = \frac{\sum_{i \in OIL, j} F_{i,j,y} \times COEF_{i,j}}{\sum_{i,j} F_{i,j,y} \times COEF_{i,j}} \quad (3)$$

$$\lambda_{Gas} = \frac{\sum_{i \in GAS, j} F_{i,j,y} \times COEF_{i,j}}{\sum_{i,j} F_{i,j,y} \times COEF_{i,j}} \quad (4)$$

Where:

$F_{i,j,y}$: the consumption of fuel i for province j in year y (tce);

$COEF_{i,j,y}$: is the emission factor (tCO₂/tce) of fuel i, taking into account the carbon content of fuel i and the percentage of oxidation of the fuel in year y

COAL, OIL and GAS respectively refers to the group of solid, liquid, and gas fuels.

Based on China Energy Statistical Yearbook 2005, the calculation of the weights of solid, liquid, and gas fuels in Southern China Power Grid are:

$$\lambda_{Coal} = 89.48\%, \quad \lambda_{Oil} = 10.24\%, \quad \lambda_{Gas} = 0.28\%$$



Step 2: Calculation of Emission Factor of Relevant Thermal Power

$$EF_{Thermal} = \lambda_{Coal} \times EF_{Coal,Adv} + \lambda_{Oil} \times EF_{Oil,Adv} + \lambda_{Gas} \times EF_{Gas,Adv} \quad (5)$$

Where: $EF_{Coal,Adv}$, $EF_{Oil,Adv}$ and $EF_{Gas, Adv}$ respectively refers to the emission factor representing best technology commercially available for fuel of coal, oil or gas fired power plants. For specific workings, see the following:

Table A11. Emission factor representing best technology commercially available for fuel of coal, oil or gas fired power plants

	Variable	Efficiency of Power Supply	Emission Coefficient of Fuel (tc/TJ)	Oxidation Rate	Emissions (tCO ₂ /MWh)
		A	B	C	D=3.6/A/1000*B*C*44/12
Coal-fired Power Plant	$EF_{Coal,Adv}$	35.82%	25.8	1	0.9508
Gas-fired Power Plant	$EF_{Gas,Adv}$	47.67%	15.3	1	0.4237
Oil-fired Power Plant	$EF_{Oil,Adv}$	47.67%	21.1	1	0.5843

$$EF_{Thermal} = \lambda_{Coal} \times EF_{Coal,Adv} + \lambda_{Oil} \times EF_{Oil,Adv} + \lambda_{Gas} \times EF_{Gas,Adv} = 0.9117 \text{ (tCO}_2\text{/MWh)}$$

Step 3: Calculation of BM of the Grid

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$$EF_{BM,y} = \frac{CAP_{Thermal}}{CAP_{Total}} \times EF_{Thermal} , \quad (6)$$

Where: CAP_{Total} is the total of new capacity additions, and $CAP_{Thermal}$ is the new capacity addition of thermal power.

Table A12. Installed Capacity of Southern China Power Grid in 2005

Installed Capacity	Unit	Guangdong	Guangxi	Yunnan	Guizhou	Total
Thermal Power	MW	35182.6	4931.2	4758.4	9634.8	54507
Hydro Power	MW	9035.7	6085.3	7993.1	7233	30347.1
Nuclear Power	MW	3780	0	0	0	3780
Wind Power and Others	MW	83.4	0	0	0	83.4
Total	MW	48081.7	11016.5	12751.5	16867.8	88717.5

Source: China Electric Power Yearbook 2006

Table A13. Installed Capacity of Southern China Power Grid in 2004

Installed Capacity	Unit	Guangdong	Guangxi	Yunnan	Guizhou	Total
Thermal Power	MW	30172.9	4378.1	4306.9	7801.8	46659.7
Hydro Power	MW	8584.6	5040.4	7058.6	6896.5	27580.1
Nuclear Power	MW	3780	0	0	0	3780

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Wind Power and Others	MW	83.4	0	0	0	83.4
Total	MW	42621	9418.5	11365.5	14698.3	78103.3

Source: China Electric Power Yearbook 2005

Table A14. Installed Capacity of Southern China Power Grid in 2003

Installed Capacity	Unit	Guangdong	Guangxi	Yunnan	Guizhou	Tianshengqiao	Total
Thermal Power	MW	27231.4	3190.1	3556.8	6465.8	0	40444.1
Hydro Power	MW	8107.2	4525.2	6543.2	3713.7	2520	25409.3
Nuclear Power	MW	3780	0	0	0	0	3780
Wind Power and Others	MW	83.4	0	0	0	0	83.4
Total	MW	39202	7715.3	10100	10179.5	2520	69716.8

Source: China Electric Power Yearbook 2004

**Table A15. Calculation of BM of Southern China Power Grid**

	Installation in year 2003	Installation in year 2004	Installation in year 2005	New Additions from 2003 to 2005	Ratio in New Additions
	A	B	C	D=C-A	
Thermal Power (MW)	40444.1	46659.7	54507	14062.9	74.01%
Hydro Power (MW)	25409.3	27580.1	30347.1	4937.8	25.99%
Nuclear Power (MW)	3780	3780	3780	0	0.00%
Wind Power (MW)	83.4	83.4	83.4	0	0.00%
	69716.8	78103.3	88717.5	19000.7	100.00%
Total (MW)	78.58%	88.04%	100%		

$$EF_{BM,y} = 0.9117 \times 74.01\% = 0.6748 \text{ tCO}_2/\text{MWh}.$$

Table A16. Baseline Emission Factor of Southern China Power Grid

	Parameter	Unit	Amount
A	Operating Margin Emission Factor	tCO ₂ /MWh	1.0119
B	Build Margin Emission Factor	tCO ₂ /MWh	0.6748
C	Combined Emission Factor (C=0.75*A+0.25*B)	tCO ₂ /MWh	0.9276

Table A17. Electricity Generation Baseline Emissions

	Parameter	Unit	Amount	Source or Equation
A	Project installed capacity	MW	23.25	See Footer7
B	Annual electricity supplied	MWh	57,595	See Footer7
C	Baseline Emissions Factor	tCO ₂ /MWh	0.9276	Table
D	Electricity generation baseline emissions	tCO ₂ /year	53,425	D= B * C



Annex 4

MONITORING PLAN

Please refer to B 7.2 in the PDD